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An Efficient Token Compression Framework for Visual Object Tracking

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Abstract

Refining visual representations by eliminating their internal feature-level redundancy is crucial for simultaneously optimizing the performance and computational cost of models in visual tracking. To enhance their performance, many contemporary Transformer-based trackers leverage a larger number of historical template frames to capture richer spatio-temporal cues. However, this strategy leads to a massive number of input visual tokens. This creates two critical issues: it imposes a quadratic computational burden and can also degrade the tracker's overall performance. To bridge this gap, we propose a ***compress-then-interact*** *tracking framework,* ***ETCTrack****, that learns to efficiently compress template tokens from historical template frames into a robust target representation, moving beyond handcrafted rules. Our method first employs the Adaptive Token Compressor to dynamically construct compact yet highly discriminative template tokens by filtering out redundant visual tokens. These refined template tokens are then processed by our Hierarchical Interaction Encoder to achieve a deep, adaptive interaction with the search features. Refined search features ensure subsequent precise target localization. Experiments on seven benchmarks demonstrate that our method outperforms current state-of-the-art trackers. ETCTrack-B224 reduces the number of template tokens by 60%, leading to a 21.4% reduction in MACs with only a 0.4% drop in accuracy. The source code are available at https://github.com/PJD-WJ/ETCTrack.*

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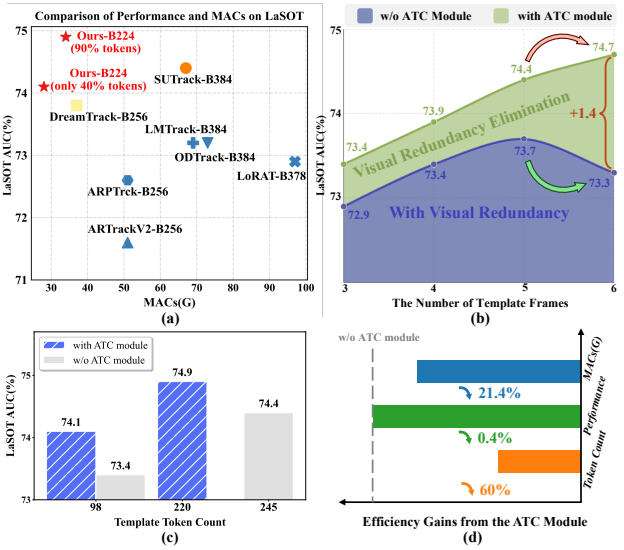


Figure 1. (a) Comparison of AUC and MACs of recent SOTA trackers. (b) Baseline models decline after the 5th frame due to visual redundancy on LaSOT, but our ATC module overcomes this, enabling sustained performance gains with more frames. (c) Impact of ATC module and template token count on LaSOT AUC(%). (d) Efficiency Gains from the ATC module.

1. Introduction

Visual tracking aims to locate a target throughout a video sequence using its initial state from the first frame as reference. Siamese network-based trackers [3, 29, 52] rose to prominence by reframing tracking as an efficient similarity-learning task between a template and a search region,

achieving a strong balance of speed and accuracy. However, their similarity-matching design revealed limitations in handling complex appearance changes and contextual interactions, motivating the development of models with richer representations and stronger relational reasoning.

This pursuit led to the successful adaptation of Transformer architectures [10, 54, 67] for visual object tracking, leveraging their powerful self-attention mechanisms to model global contextual information and complex feature interactions more effectively. Key methods such as those in [7, 8, 63, 65, 69] quickly established the prowess of this paradigm, setting new state-of-the-art performance benchmarks. To further improve robustness against significant target appearance variations and for longe-term tracking, the method of multi-frame visual tracking emerged. This approach, exemplified by methods [9, 19, 58, 68], while aiming to build richer and more robust target representations from historical template frames, simultaneously introduces a critical new challenge: **a massive number of input visual tokens, which may introduce substantial visual redundancy that degrades model performance and incurs additional computational cost.**

Inspired by the core principle from Multimodal Large Language Models (MLLMs) [26, 41, 43, 44, 70] that mitigating visual redundancy is fundamental to balancing model capability with computational cost , we argue that this principle directly applies to a parallel challenge in multi-frame visual tracking. To empirically validate this assertion, we conducted a preliminary study, with results shown in Figure.1(b). For this study, we use the same number of template frames for both training and inference. We observe that when our representative baseline tracker, OSTrack [65] equipped with a novel backbone [53], is augmented with an increasing number of historical template frames (**blue line**), its performance begins to decline significantly after the 5th frame. However, this performance ceiling is overcome when our ATC module (discussed further below) is introduced to eliminate the visual redundancy (**green line**). **Our experiment confirms that visual redundancy is a critical bottleneck in multi-frame visual tracking.**

To alleviate the above problem, we propose a compress-then-interact framework, named **ETCTrack**, which effectively compresses visual tokens from historical template frames to eliminate visual redundancy and enable more effective context modeling. The core of this framework consists of two components designed to work in synergy. (1) The Adaptive Token Compressor (ATC), a learnable module that acts as a crucial pre-processing stage before features enter our Hierarchical Interaction Encoder. It moves beyond handcrafted selection rules, which often rely on predefined metrics like attention maps or fixed spatial rules to measure token importance. Instead, our ATC module learns to dynamically evaluate the contextual importance of each token from historical template frames, guided directly by the final tracking objective. This allows it to construct a compact yet powerful subset of tokens that preserves the target's most discriminative features, providing a robust representation for interaction while reducing the computational cost (MACs). (2) Our Hierarchical Interaction Encoder, constructed by stacking several Hierarchical Interaction Block (HIBlock) modules, then leverages compressed template representation to deeply interact with the search features. This is achieved through a multi-stage interaction process within each block, involving an context-aware enrichment of the template, followed by unified feature learning, and a final template-guided refinement of the search features. Finally, the interaction-enhanced search features are processed by the prediction head to estimate the target's state. As shown in Figure 1(a)(c)(d), our method achieves highest performance with the **lowest computational cost** among leading trackers, thanks to our core components. To summarize, the main contributions of this work are as follows:

- We propose a novel compress-then-interact framework for visual tracking that resolves the critical trade-off between performance and efficiency in multi-frame context modeling, establishing a new SOTA by outperforming leading trackers in both accuracy and computational cost.
- We introduce two key modules: the ATC module for efficiently compressing redundant tokens while preserving discriminative features, and the HIBlock for deep and effective contextual information interaction.
- Our method achieves SOTA results on seven challenging visual tracking benchmarks, including GOT10K, LaSOT, $\text{LaSOT}_{\text{ext}}$, TrackingNet, NfS, TNL2K, and OTB100.

2. Related Work

2.1. Visual Tracking Architectures

The advent of deep learning has revolutionized visual tracking, with the Siamese tracking framework [3] emerging as a dominant architecture. Its compelling balance of accuracy and real-time efficiency spurred numerous works that aimed to significantly enhance overall tracking capabilities and robustness [11, 14, 30, 31]. In recent years, the field has decisively shifted towards Transformer-based architectures [53, 54, 67], valued for their superior capabilities in modeling global context and intricate feature interactions. This paradigm has evolved rapidly, from early hybrid models that integrated Transformer modules with CNNs [8, 15, 16, 63], to more recent unified, one-stream architectures that handle both feature extraction and interaction [9, 61, 62, 65]. Concurrently, the adoption of advanced hierarchical backbones like the Swin Transformer [37, 42, 50] has further pushed performance boundaries, shaping the diverse landscape of modern state-of-the-art trackers [7, 21–24, 33, 51].

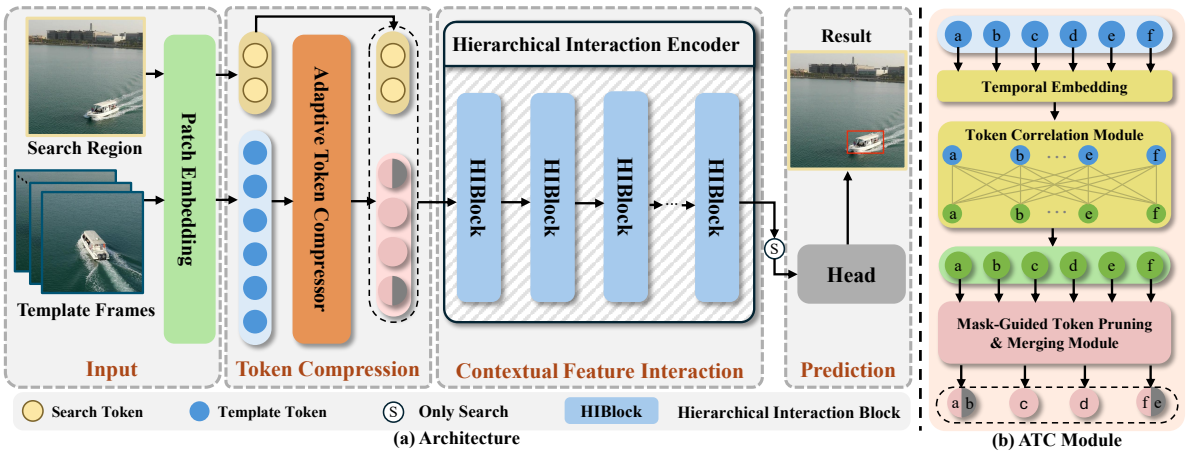


Figure 2. (a) **ETCTrack Framework Architecture.** The process begins with our ATC module, which effectively compresses visual tokens from historical template frames to eliminate visual redundancy. These compressed tokens, along with search region tokens, are then fed into the Hierarchical Interaction Encoder for contextual feature interaction. Finally, the enhanced search features are sent to the Prediction Head to predict bounding box. (b) The detailed architecture of our core ATC module.

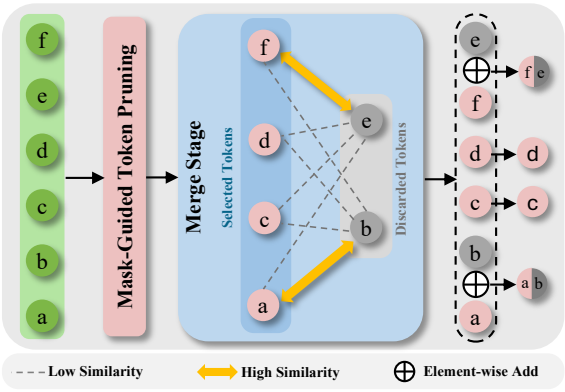


Figure 3. The Mask-Guided Token Pruning and Merging Module.

2.2. Temporal Modeling in Visual Tracking

Trackers have explored several strategies to better model temporal information. One such approach incorporates motion priors, using autoregressive models to predict locations from historical trajectories [2, 56], but is prone to error accumulation during occlusions. Another strategy focuses on adaptive appearance modeling, where methods like [49, 60] learn to evolve the target's representation, replacing handcrafted update rules. The most prominent current approach, leveraged by advanced trackers [5, 19, 68], is using historical template frames for a richer, more comprehensive spatio-temporal context. However, the increasing number of input visual tokens often imposes a quadratic computational burden and can also degrade tracking performance due to visual redundancy.

2.3. Visual Token Compression

The challenge of processing long video sequences with high visual redundancy has become a central research topic in Multimodal Large Language Models (MLLMs) [1, 20, 34, 36, 40]. The massive number of visual tokens from video inputs, coupled with the quadratic complexity of Transformers, has spurred the development of various visual token compression techniques, such as those based on handcrafted rules [6, 32, 48, 66]. While these rule-based solutions offer a straightforward approach, they are often suboptimal as they rely on predefined criteria, such as importance scores derived from attention maps or fixed spatial rules, to evaluate token importance. Such non-learnable criteria may not align with the final tracking objective and can risk discarding crucial information. Therefore, this work focuses on developing an adaptive, learning-based framework to eliminate visual redundancy in tracking, aiming to simultaneously reduce computational cost and improve model performance.

3. Our Method

This section details the architecture of our proposed visual tracking framework, ETCTrack. We first provide an overview of the overall pipeline, then elaborate on its three core components: the backbone, the ATC module, and the HIBlock. Finally, we describe the prediction head and the final training objective.

3.1. Overview

The framework of the ETCTrack is demonstrated in Figure.2(a). The core components of our framework include an ATC module for eliminating visual redundancy, a HIBlock

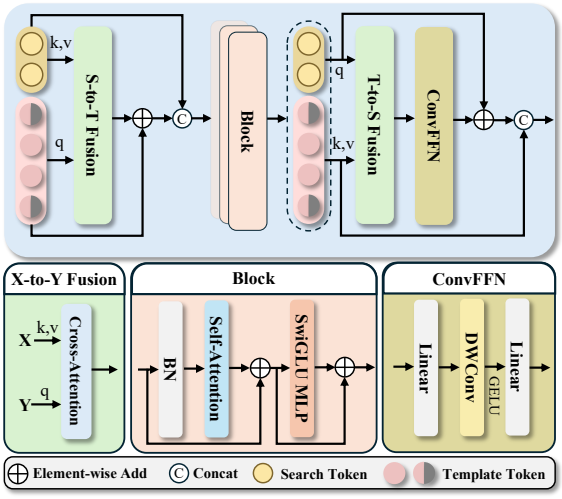


Figure 4. The structure of the Hierarchical Interaction Block.

for contextual feature interaction, and a prediction head. At each tracking step, the framework processes historical template frames and the current search frame. These inputs are first partitioned into patches and embedded to generate template and search tokens. This collection of template tokens is then fed into our ATC module, which dynamically constructs a powerful subset by filtering out redundant or noisy information. This refined template tokens, along with the search tokens, is subsequently passed to Hierarchical Interaction Encoder for contextual feature interaction, where a multi-stage interaction process robustly correlates the target's essential features with search features. Finally, the resulting interaction-enhanced search tokens are fed into a prediction head to estimate the target's state.

3.2. Adaptive Token Compressor

The ATC module is a core component of our framework, designed to eliminate visual redundancy in multi-frame tracking. Inspired by similar challenges in Multimodal Large Language Models (MLLMs) [1, 34, 70], the ATC module moves beyond handcrafted rules by learning to intelligently construct a compact yet highly representative subset of tokens from historical template frames. It employs a powerful global attention mechanism to evaluate the contextual importance of each token, allowing it to filter out noisy and redundant information. This process yields a highly discriminative template representation that simultaneously enhances tracking robustness while reducing computational cost.

As shown in Figure 2(b), the module takes template features $\boldsymbol{F}_{\text{z}} \in \mathbb{R}^{(T\cdot L)\times C}$ as input. First, we restore the explicit spatio-temporal structure by reshaping the input into $\boldsymbol{Z}_{\text{p}} \in \mathbb{R}^{T\times L\times C}$, where T is the number of historical template frames, L is the number of tokens per frame, and C is the feature dimension. To distinguish tokens from different frames, a learnable temporal positional embedding $\boldsymbol{E}_{\text{temp}} \in \mathbb{R}^{T\times 1\times C}$ is then added to $\boldsymbol{Z}_{\text{p}}$. The resulting vector is flattened along its temporal and spatial dimensions to form $\boldsymbol{Z}_{\text{temp}} \in \mathbb{R}^{(T\cdot L)\times C}$. Next, $\boldsymbol{Z}_{\text{temp}}$ is processed by our proposed Token Correlation Module (TCM), which consists of N_{atc} stacked self-attention layers. The TCM, through its global self-attention mechanism, deeply interacts with all tokens from template frames, which already have temporal positional information embedded. This allows the TCM to model the comprehensive spatio-temporal relationships across the template frames, enabling it to understand complex correlations between all template tokens and identify the most discriminative and informative parts. This process yields the contextualized feature representations $\boldsymbol{Z}_{\text{context}}$:

$$\boldsymbol{Z}_{\text{context}} = \text{TCM}(\boldsymbol{Z}_{\text{temp}}) \in \mathbb{R}^{(T\cdot L)\times C} \tag{1}$$

After obtaining $\boldsymbol{Z}_{\text{context}}$, we introduce the Mask-Guided Token Pruning module to evaluate token significance, as illustrated in Figure 3. We employ a fixed random projection to generate token importance scores $\boldsymbol{S}$, acting as a randomized token selection mechanism to enhance the model's capability in learning efficient similarity-based merging. Based on the descending order of $\boldsymbol{S}$, the tokens are partitioned into a preserved target set $\boldsymbol{\mathcal{A}} = \{\boldsymbol{a}_j\}_{j=1}^{K_{\text{target}}}$ (high score) and a redundant source set $\boldsymbol{\mathcal{B}} = \{\boldsymbol{b}_i\}_{i=1}^{K_{\text{merge}}}$ (low score). The number of merged tokens is determined by a predefined keep rate $r \in (0, 1)$, yielding $K_{\text{target}} = \lfloor r \cdot (T \cdot L) \rfloor$ and $K_{\text{merge}} = (T \cdot L) - K_{\text{target}}$.

To prevent semantic information loss from directly discarding $\boldsymbol{\mathcal{B}}$, we propose a Guided Similarity Merging strategy. Each source token $\boldsymbol{b}_i \in \boldsymbol{\mathcal{B}}$ is absorbed into its most semantically similar target token $\boldsymbol{a}_j \in \boldsymbol{\mathcal{A}}$ via greedy cosine similarity matching:

$$\boldsymbol{a}'_j = \boldsymbol{a}_j + \sum_{i\in\Omega_j} \boldsymbol{b}_i \tag{2}$$

where $\Omega_j = \left\{i \mid j = \arg\max_k (\boldsymbol{b}_i \cdot \boldsymbol{a}_k)/(\|\boldsymbol{b}_i\|_2 \|\boldsymbol{a}_k\|_2)\right\}$ denotes the indices of source tokens assigned to $\boldsymbol{a}_j$. The updated feature set $\boldsymbol{F}_{\text{comp}} = \{\boldsymbol{a}'_j\}_{j=1}^{K_{\text{target}}} \in \mathbb{R}^{K_{\text{target}}\times C}$ effectively eliminates redundancy while yielding a highly efficient, context-aware template representation.

3.3. Hierarchical Interaction Encoder

Backbone. The backbone architecture for state-of-the-art visual trackers [61, 68], has increasingly shifted towards ViT [10] for their powerful feature extraction capabilities. First, fine-grained recognition tasks like visual tracking require hierarchical features to capture information at various scales. However, the majority of standard pre-trained models are based on plain ViT architectures that produce a single-scale, non-hierarchical feature map, which is suboptimal for robustly localizing targets of different sizes. To address this, Fast-iTPN [53] was chosen as our backbone.

We take historical template frames $\boldsymbol{Z} \in \mathbb{R}^{T\times 3\times H_z\times W_z}$ and a search region $\boldsymbol{X} \in \mathbb{R}^{3\times H_x\times W_x}$ as inputs. Both $\boldsymbol{Z}$ and $\boldsymbol{X}$ are first downsampled using convolutional layers with a stride of 4. The downsampled features are then passed through MLP layers and two convolutional merging layers, which segment the inputs into non-overlapping patches. This process yields patch embeddings for the historical template frames and the search region, denoted as $\boldsymbol{F}_z \in \mathbb{R}^{N_z\times C}$ and $\boldsymbol{F}_x \in \mathbb{R}^{N_x\times C}$, respectively, where $N_z = \frac{T\cdot H_z\cdot W_z}{16^2}$, $N_x = \frac{H_x\cdot W_x}{16^2}$. The resulting patches are then concatenated along the spatial dimension to form $\boldsymbol{F}_{zx} \in \mathbb{R}^{L\times C}$, where $L = N_z + N_x$. $\boldsymbol{F}_{zx}$ subsequently serves as the input to our core Hierarchical Interaction Encoder, where template and search tokens undergo a deep, multi-stage interaction process.

Hierarchical Interaction Block. The principle of jointly performing feature extraction and relation modeling within a backbone, as demonstrated by effective tracker[65], has proven to be a powerful approach for visual tracking. By processing template and search region tokens together, these methods enable rich feature interaction. However, the extent and nature of this interaction can be further deepened and made more explicit. While joint self-attention allows for implicit correlation, a more structured, multi-stage interaction mechanism is beneficial to fully allow the template representation to guide the search process and, in turn, for the search context to refine the template representation. This motivates the design of our HIBlock.

The detailed structure of the HIBlock is shown in Figure 4. The HIBlock is the central core component of ETCTrack. It takes the compressed template tokens from the ATC module and the search tokens as input. The HIBlock is designed to perform a deep, asymmetric interaction process. This hierarchical and multi-stage design ensures a highly adaptive exchange of information between the template tokens and the search tokens.

The detailed workflow of the HIBlock is as follows. For a single sample, the block receives the compressed template tokens $\boldsymbol{F}_{\text{comp}} \in \mathbb{R}^{K\times C}$ and the search tokens $\boldsymbol{F}_{\text{x}} \in \mathbb{R}^{N_x\times C}$. First, to make the template representation aware of the current search context, an cross-attention fusion is performed where the template tokens $\boldsymbol{F}_{\text{comp}}$ act as Q, while the search tokens $\boldsymbol{F}_{\text{x}}$ serve as K and V. This step yields the context-aware template tokens, denoted as $\boldsymbol{F}'_{\text{comp}}$. These are then concatenated with the original search tokens and fed into a stack of M standard backbone blocks [53], for deep, joint feature modeling. The output of this stage, $\boldsymbol{F}_{\text{block}}$ is subsequently split back into its template and search tokens, yielding deeply encoded features $\boldsymbol{F}''_{\text{comp}}$ and $\boldsymbol{F}'_{\text{x}}$, respectively. Next, to explicitly guide the search process, a second cross-attention is performed where the search tokens $\boldsymbol{F}'_{\text{x}}$ query the deeply encoded template tokens $\boldsymbol{F}''_{\text{comp}}$, resulting in the enhanced search tokens $\boldsymbol{F}''_{\text{x}}$. Finally, these enhanced features

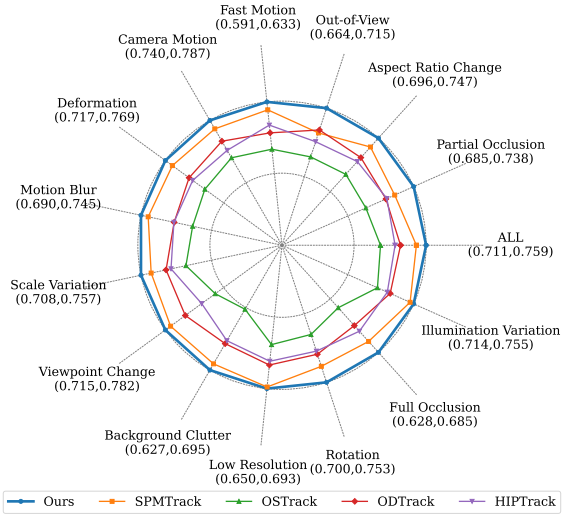


Figure 5. AUC scores of different attributes on LaSOT [12].

are passed through a Convolutional Feed-Forward Network (ConvFFN) for final refinement, producing the output features $\boldsymbol{F}_{\text{out}} \in \mathbb{R}^{N_x\times C}$ which are then sent to the prediction head. The process is summarized as follows:

$$\begin{aligned}
\boldsymbol{F}'_{\text{comp}} &= \boldsymbol{F}_{\text{comp}} + \text{CrossAttention}(\boldsymbol{F}_{\text{comp}}, \boldsymbol{F}_{\text{x}}, \boldsymbol{F}_{\text{x}}),\\
\boldsymbol{F}_{\text{block}} &= \text{BackboneBlocks}(\text{Concat}(\boldsymbol{F}'_{\text{comp}}, \boldsymbol{F}_{\text{x}})),\\
\boldsymbol{F}''_{\text{x}} &= \boldsymbol{F}'_{\text{x}} + \text{CrossAttention}(\boldsymbol{F}'_{\text{x}}, \boldsymbol{F}''_{\text{comp}}, \boldsymbol{F}''_{\text{comp}}),\\
\boldsymbol{F}_{\text{out}} &= \text{ConvFFN}(\boldsymbol{F}''_{\text{x}}).
\end{aligned} \tag{3}$$

3.4. Head and Loss

The output features $\boldsymbol{F}_{\text{out}}$ from the Hierarchical Interaction Encoder are forwarded to a Fully Convolutional Network (FCN), which comprises L stacked Conv-BN-ReLU layers in each output branch. The FCN generates three types of outputs: a target classification score map of size $\mathbb{R}^{\frac{H_x}{P}\times\frac{W_x}{P}}$, an offset map of size $\mathbb{R}^{2\times\frac{H_x}{P}\times\frac{W_x}{P}}$ to compensate for discretization errors introduced by downsampling, and normalized bounding box dimensions of size $\mathbb{R}^{2\times\frac{H_x}{P}\times\frac{W_x}{P}}$.

During the training phase, the classification and regression objectives are jointly optimized. For the classification task, which aims to distinguish the target from the background, we adopt the weighted focal loss [39]. For bounding box regression, which is crucial for precise localization, we employ a combination of the L_1 loss and the generalized IoU loss [47] on the predicted box coordinates. The comprehensive optimization objective is formulated as:

$$L = L_{\text{cls}} + \lambda_{\text{iou}} L_{\text{iou}} + \lambda_{L_1} L_1, \tag{4}$$

where the weights λ_{L_1} and λ_{iou} are set to 5 and 2.

Method	Source	GOT-10k*			LaSOT			LaSOT$_{ext}$			TrackingNet			TNL2k	
		AO	SR$_{0.5}$	SR$_{0.75}$	AUC	P$_{norm}$	P	AUC	P$_{norm}$	P	AUC	P$_{norm}$	P	AUC	P
ETCTrack-B224	Ours	**79.2**	**90.2**	78.6	**74.9**	**85.1**	**82.7**	**54.6**	**65.9**	**62.4**	86.0	90.6	85.6	**61.3**	**65.9**
ETCTrack-S224	Ours	75.8	86.6	75.0	73.4	83.5	81.1	53.7	65.0	61.1	85.4	90.1	84.8	60.0	63.4
ETCTrack-T224	Ours	74.9	85.6	72.6	71.8	82.1	78.9	52.5	63.9	59.4	84.6	89.3	83.2	58.3	61.0
DreamTrack-B256 [17]	CVPR25	77.5	87.1	74.2	73.8	83.4	80.6	53.1	64.1	59.8	85.8	90.0	85.3	60.4	63.2
ARPTrack-B256 [35]	CVPR25	77.7	87.3	74.3	72.6	81.4	78.5	52.0	62.9	58.7	85.5	90.0	85.3	-	-
MCITrack-B224†[27]	AAAI25	77.9	87.5	**78.8**	74.0	83.6	81.4	52.0	62.5	59.0	**86.4**	**91.0**	**86.3**	60.3	64.4
TemTrack-B256 [60]	AAAI25	74.9	84.8	71.7	72.0	82.1	79.1	52.4	63.3	60.2	84.3	88.8	83.5	58.8	-
ARTrackV2-B256 [2]	CVPR24	75.9	85.4	72.7	71.6	80.2	77.2	50.8	61.9	57.7	84.9	89.3	84.5	59.2	-
VideoTrack[59]	CVPR23	72.9	81.9	69.8	70.2	-	76.4	-	-	-	83.8	88.7	83.1	-	-
MixFormer-22k[9]	CVPR22	70.7	80.0	67.8	69.2	78.7	74.7	-	-	-	83.1	88.1	81.6	-	-
OSTrack-256[65]	ECCV22	71.0	80.4	68.2	69.1	78.7	75.2	47.4	57.3	53.3	83.1	87.8	82.0	54.3	-
TransT [8]	CVPR21	67.1	76.8	60.9	64.9	73.8	69.0	-	-	-	81.4	86.7	80.3	50.7	51.7
SiamRPN++[31]	CVPR19	51.7	61.6	32.5	49.6	56.9	49.1	34.0	41.6	39.6	73.3	80.0	69.4	41.3	41.2
SiamFC [3]	ECCVW16	34.8	35.3	9.8	33.6	42.0	33.9	23.0	31.1	26.9	-	-	-	29.5	28.6
Some Trackers with Higher Resolution															
OSTrack-384[65]	ECCV22	73.7	83.2	70.8	71.1	81.1	77.6	50.5	61.3	57.6	83.9	88.5	83.2	55.9	-
F-BDMTrack-384[64]	ICCV23	75.4	84.3	72.9	72.0	81.5	77.7	50.8	61.3	57.8	84.5	89.0	84.0	57.8	59.4
LoRAT-B378[4]	ECCV24	73.7	82.6	72.9	72.9	81.9	79.1	53.1	64.8	60.6	84.2	88.4	83.0	59.9	63.7
HIPTrack-B384[4]	CVPR24	77.4	88.0	74.5	72.7	82.9	79.5	53.0	64.3	60.6	84.5	89.1	83.8	-	-
ARTrackV2-B384 [2]	CVPR24	77.5	86.0	75.5	73.0	82.0	79.6	52.9	63.4	59.1	85.7	89.8	85.5	-	-
ODTrack-B384 [68]	AAAI24	77.0	87.9	75.1	73.2	83.2	80.6	52.4	63.9	60.1	85.1	90.1	84.9	60.9	-
TemTrack-B384 [60]	AAAI25	76.1	84.9	74.4	73.1	83.0	80.7	53.4	64.8	61.0	85.0	89.3	84.8	-	-
SPMTrack-B384 [5]	CVPR25	76.5	85.9	76.3	74.9	84.0	81.7	-	-	-	86.1	90.2	85.6	62.0	66.7
DreamTrack-B384 [17]	CVPR25	78.3	87.9	76.6	75.0	84.2	81.7	54.5	65.3	61.1	86.5	90.6	85.9	61.2	67.4
ETCTrack-B384	Ours	**80.1**	**89.5**	**80.7**	**75.9**	**85.9**	**84.0**	**55.1**	**66.2**	**62.5**	**87.3**	**91.7**	**88.2**	**63.0**	**68.4**

Table 1. Performance comparison with SOTA trackers on the test sets of GOT-10k [25], LaSOT [12], LaSOT$_{\text{ext}}$ [13], TrackingNet [45], and TNL2K [55]. For GOT-10k, an asterisk (*) denotes models trained exclusively on its training set. MCITrack-B224† was locally trained under identical model, GPU, and configuration conditions. The top two results are highlighted with **bold** and underlined fonts, respectively.

	DreamTrack-B384	SPMTrack-B384	ARTrackV2-L384	HIPTrack-B384	ODTrack-B384	LoRAT-L378	SeqTrack-B384	**ETCTrack-B224**	**ETCTrack-B384**
NfS	66.1	67.4	68.4	68.1	-	66.7	66.7	**71.3**	71.2
OTB100	72.0	72.7	-	71.0	72.3	-	-	73.3	**73.6**

Table 2. Comparison with state-of-the-art methods on NfS [28] and OTB100 [57] benchmarks in AUC score. The top two results are highlighted with **bold** and underlined fonts, respectively.

#	ATC Module	HIBlock	LaSOT		TNL2K		FLOPs (G)
			AUC	P$_{norm}$	AUC	P$_{norm}$	
1	✗	✗	73.7	83.7	59.9	77.7	33
2	✓	✗	74.4	84.6	60.6	78.3	32
3	✗	✓	74.4	84.5	60.5	78.0	36
4	✓	✓	**74.9**	**85.1**	**61.3**	**78.8**	34

Table 3. Ablation studies on ATC module and HIBlock.

Keep Ratio(r)	LaSOT↑	LaSOT$_{\text{ext}}$↑	NfS↑	TNL2K↑	MACs(G)↓
1.0 (w/o ATC)	74.4	53.5	71.2	60.5	35.9
0.9	**74.9**(+0.5)	**54.6**(+1.1)	**71.3**(+0.1)	**61.3**(+0.8)	34.1
0.7	74.5(+0.1)	53.8(+0.3)	71.1(-0.1)	60.8(+0.3)	31.7
0.5	74.3(-0.1)	54.0(+0.5)	71.3(+0.1)	60.6(+0.1)	**28.2**

Table 4. Performance vs. keep ratio (r).

4. Experiment

4.1. Implementation Details

Training. We use Fast-iTPN [53] model as our backbone. The training data includes LaSOT [12], GOT-10k [25], TrackingNet [45], COCO [38] and VastTrack [46]. We employ the AdamW to optimize the network parameters with initial learning rate of 2×10^{-5} for the backbone, 2×10^{-4} for the rest, and set the weight decay to 10^{-4}. We set the training epochs to 300 epochs. 60,000 search images are randomly sampled in each epoch. The learning rate drops by a factor of 10 after 240 epochs. The model is conducted on a server with four 80GB Tesla A800 GPUs, using a batch size of 16, where each batch consists of 2 search images and 5 template images.

Inference. During inference, our core strategy is to compress template information using the ATC module, which retains only the most representative tokens based on a token keep ratio(r) of 0.9 to create a potent yet efficient representation. Consistent with our training process, this mechanism is applied to 5 historical template frames that are processed

TCM Architecture	AUC	Speed(fps)
ViT-B(MAE pre-trained, 12 layers)	74.7	44
ViT-B(from scratch, 12 layers)	74.2	44
Transformer Encoder(8 layers)	74.9	46
Fast-ITPN(The first 8 layers)	**74.9**	**48**

Table 5. Ablation on Different TCM Architecture on LaSOT.

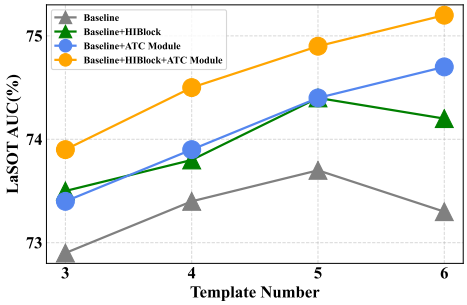


Figure 6. LaSOT AUC vs. template frames for our variants.

alongside the current search region. Furthermore, to effectively mitigate error accumulation in long-term tracking, we employ a dynamic memory bank that is selectively updated. A new frame is added only if its maximum classification score surpasses a confidence threshold τ, ensuring the bank stores only reliable, high-quality historical template frames for robust long-term tracking.

4.2. Comparison with State-of-the-Art Trackers

To demonstrate the effectiveness of ETCTrack, we compare it against state-of-the-art trackers on seven diverse benchmarks, including GOT-10k, TrackingNet, LaSOT, LaSOT$_{ext}$, TNL2K, NfS, and OTB100.

GOT-10k [25]. GOT-10K. The GOT-10k benchmark is a large-scale tracking dataset comprising over 10,000 video sequences. Its strict protocol mandates that trackers use only its designated training set for training, a guideline we adhered to for our framework. As shown in Table 1, our ETCTrack-B224 achieved an AO score of 79.2%, surpassing the second-place MCITrack-B224 (77.9% AO). Our proposed method significantly outperforms previous SOTA trackers, even achieving higher performance than all higher-resolution trackers.

LaSOT [12]. LaSOT is a large-scale, long-term tracking benchmark that includes 1,120 sequences for training and 280 for testing, known for its challenging scenarios. As presented in Table 1 and Figure 5, our ETCTrack-B384 achieves a new state-of-the-art result on this demanding benchmark. Specifically, our method surpasses the strong competitor DreamTrack-B384 by 0.9%, 1.7%, and 2.1% in terms of AUC, P_{norm}, and P scores, respectively.

LaSOT$_{ext}$ [13]. This benchmark is an extension of LaSOT, introducing an additional 150 long-term sequences with numerous challenges, such as fast-moving small objects. As reported in Table 1, our method outperforms other trackers by a substantial margin, achieving the highest scores across all metrics (AUC, P_{norm}, and P). For instance, our method achieves an AUC of 55.1%, surpassing the previous best, DreamTracker-B384, by 0.6%.

Number of Templates	AUC	P$_{norm}$	P
3 frames	73.9	83.9	81.5
4 frames	74.5	84.4	81.8
5 frames	74.9	85.1	82.7
6 frames	75.2	85.3	83.0

Table 6. Ablation on the Number of Templates on LaSOT.

TrackingNet [45]. TrackingNet is a large-scale tracking dataset with over 30,000 training sequences and 511 test sequences. This benchmark focuses on challenges encountered when tracking objects in the wild, such as background clutter, full occlusion, and low resolution. The results on the TrackingNet test set are shown in Table 1. Our ETCTrack achieves an AUC score of 87.3%, outperforming the strong runner-up DreamTrack-B384 (86.5% AUC) and demonstrating the robustness of our method in unconstrained environments.

TNL2K [55]. TNL2K is a recently released large-scale tracking dataset containing 700 challenging test sequences, known for its diverse and difficult scenarios. On this demanding benchmark, our tracker again achieves state-of-the-art performance, as shown in Table 1. Notably, our ETCTrack-B384 obtains an AUC of 63.0%, surpassing strong competitors like SPMTrack-B384 (62.0% AUC) by a significant margin.

OTB100 [57] and NfS [28]. OTB100 is a popular short-term tracking benchmark with 100 sequences covering 11 challenges like deformation and occlusion, while NfS contains 100 challenging, high-frame-rate videos. As shown in Table 2, our ETCTrack-B224 surpasses all state-of-the-art trackers with significantly larger model sizes, demonstrating the superior efficiency and effectiveness of our design.

4.3. Ablation Study and Analysis

In the ablation study, we will investigate the impact of our ATC module, HIBlock, number of template frames, different backbones, and varying token keep ratios (r) on model performance. All the ablation study is based on ETCTrack-B224. Unless otherwise specified, we will default to r value of 0.9 and 5 template frames.

The Importance of ATC Module and HIBlock. Table 3 validates the significance of the two core components in our work: the ATC module and the HIBlock. In this study, removing the ATC module means all template tokens are used without redundancy elimination, while removing the HIBlock means replacing it with the standard backbone blocks

Backbone	Our method	LaSOT		TNL2K	
		AUC	P_{norm}	AUC	P_{norm}
ViT-B	✗	69.9	79.9	58.4	76.1
	✓	73.4(+3.5)	83.8(+3.9)	60.6(+2.2)	78.1(+2.0)
Fast-iTPN-B	✗	72.8	83.1	59.4	77.0
	✓	**74.9**(+2.1)	**85.1**(+2.0)	**61.3**(+1.9)	**78.8**(+1.8)

Table 7. Backbone impact on LaSOT and TNL2K performance.

[53] for interaction. Our baseline model, where both ATC module and HIBlock are removed, processes all 5 template frames and the search region directly through the backbone before prediction. The results on LaSOT are analyzed as follows (TNL2K results in Table 3). Introducing only the ATC module to the baseline substantially improves performance by +0.7 AUC, demonstrating its high effectiveness in visual redundancy elimination. Adding only the HIBlock boosts the AUC score by +0.7, showcasing its powerful contextual modeling. When both modules are integrated, the ATC module provides a further +0.5 AUC gain on top of the HIBlock-equipped baseline, confirming the combined effect of our two components.

The Number of Templates. As shown in Table 6 and Figure.6, we investigate the impact of varying the number of template frames on LaSOT. While adding more frames initially boosts performance (+1.3% AUC gain from 3 to 6 frames), we observe that baseline models without our ATC module suffer a significant performance decline beyond 5 frames. This suggests that naively adding frames introduces detrimental redundant information. In contrast, our model with ATC avoids this issue, effectively extracting useful cues from more template frames.

Variants of the TCM Architectures. We explored several architectures for the Token Correlation Module (TCM): ViT-B (trained from scratch or fine-tuned from MAE pre-trained weights [18]), a 8-layer Transformer encoder trained from scratch, and the first 8 layers of Fast-ITPN-B fine-tuned from its pre-trained weights. As shown in Table 5, the first 8 layers of Fast-ITPN-B (without additional model parameters) surpass both ViT-B variants and an 8-layer Transformer encoder on LaSOT in terms of either tracking accuracy or inference speed. This result indicates that a larger model scale is not the decisive factor for this task; instead, a more streamlined and well-tailored architecture demonstrates superior competitiveness.

Effect of backbone. Specifically, integrating our method with either ViT-B or Fast-iTPN-B results in significant improvements on the LaSOT and TNL2K. Notably, the Fast-iTPN-B equipped with our method achieves the best performance, reaching 74.9 AUC on LaSOT, as well as 61.3 AUC on TNL2K, validating its effectiveness and adaptability.

Visualization of token elimination. Figure 7 illustrates the token reduction process across sampled temporal templates. By employing the ATC module, **60% of the total tokens** are effectively eliminated (represented by grey blocks) to prune redundant information while preserving informative features. Notably, the token elimination is more concentrated in the intermediate frames (2nd to 4th), which can be attributed to the high information redundancy following the TCM fusion; in these frames, a sparse subset of tokens is sufficient to represent the essential context. In contrast, the first and fifth frames retain more tokens, as the initial frame provides the most reliable target appearance and the final frame captures the latest dynamic cues, both of which are indispensable for robust tracking.



Figure 7. Visualization of visual redundancy elimination.

Limitation. While our approach achieves state-of-the-art performance with a very low computational cost, we believe there is a particularly exciting and impactful direction for future research. This valuable area of investigation is the development of fully dynamic token compression mechanisms, where the level of compression could adapt in real-time based on the complexity of the tracking scenario or the target's rapidly changing state.

5. Conclusion

In this work, we introduced ETCTrack, a novel compress-then-interact visual tracking framework that effectively addresses visual redundancy in multi-frame Transformer trackers. Our framework consists of two key innovations: the Adaptive Token Compressor module, which constructs a sparse, potent token set from historical template frames, and the Hierarchical Interaction Block, which performs a deep, multi-stage fusion of this compressed representation with the search region. Extensive experiments demonstrate that ETCTrack achieves state-of-the-art performance on seven challenging benchmarks while maintaining a low computational cost. We believe these results validate that explicit information condensation is a promising direction for developing future high-performance trackers.

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